CCC-C-426b
December 13, 1962

SUPERSEDING
Int. Fed. Spec. CCC-C-00426a (ARMY-QMC)
February 26, 1962 and
Fed. Spec. CCC-C-426
April 16, 1957

FEDERAL SPECIFICATION

CLOTH, COTTON, DRILL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for cloth, cotton, drill.

1.2 Classification.

1.2.1 *Types and classes.* The cloth covered by this specification shall be of the following types and classes (see 6.2):

Type I.—30 inches* 72 by 60, 2.50 yd. (Commercial designation).

Type II.—30 inches* 72 by 48, 2.50 yd. (Commercial designation).

Type III.—30 inches* 72 by 48, 2.85 yd. (Commercial designation).

Class 1.—Greige.

Class 2.—Desized and preshrunk.

Class 3.—Bleached white or dyed and preshrunk.

* Prorata widths of like construction are acceptable when specified (see 6.2).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

Federal Specifications:

V-T-276—Thread, Cotton.

CCC-T-191—Textile Test Methods.

PPP-P-51—Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks, and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications,

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Standards, and Handbooks from established distribution points in their agencies.)

Military Standard:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Rules and Regulations Under The Textile Fiber Products Identification Act.

(Copies may be obtained, free of charge, from the Federal Trade Commission, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Standard sample. When a standard sample is available, the cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material.

3.2.1 *Yarn.* The yarn shall be made from clean cotton, carded, drawn, and evenly spun into single yarn for both the warp and filling.

3.3 Color. Classes 1 and 2 shall be natural color. Class 3 cloth shall be bleached or dyed as specified (see 6.2). The use of sulfur dyes or dyes containing elementary sulfur or compounds capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed cloth shall show no more free or sulfide sulfur than the standard sample when tested as specified in 4.4.

3.3.1 *Matching.* The dyed cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

3.3.2 *Colorfastness.* The dyed and finished cloth shall show colorfastness to laundering, wet dry cleaning, perspiration, crocking, and light equal to or better than the standard sample. When no standard sample is established or when the standard sample is not a reference for color fastness, the dyed and finished cloth shall show good colorfastness to laundering, wet dry cleaning, perspiration, crocking, and light. Testing shall be as specified in 4.4.

TABLE I.—*Physical requirements*

Type	Class	Weight per sq. yard	Yarns per inch		Breaking strength (grab method)		Weave (for all types)
			Warp	Filling	Warp	Filling	
		Ounces			*Pounds*	*Pounds*	
I	1	7.7	72	60	130	90	3 harness
	2	7.5	74	58	120	85	2 left twill
	3	6.5	74	56	105	70	1
II	1	7.7	72	48	130	85	
	2	7.5	74	46	120	72	
	3	6.5	74	44	105	62	
III	1	6.7	72	48	110	65	
	2	6.5	74	46	100	60	
	3	6.0	74	44	90	55	

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3.4 Physical requirements. The finished cloth shall conform to the minimum requirements specified in table I when tested as specified in 4.4.

3.4.1 *Width.* The width shall be as specified (see 6.2) and shall be the minimum acceptable width inclusive of selvages.

3.5 Sizing, finishing, and other nonfibrous materials.

3.5.1 *Class* 1. The total sizing, finishing, and other nonfibrous materials shall not exceed 12.0 percent when tested as specified in 4.4.

3.5.2 *Class* 2. The total sizing, finishing, and other nonfibrous materials shall not exceed 5.0 percent when tested as specified in 4.4.

3.5.3 *Class* 3. The total sizing, finishing, and other nonfibrous materials shall not exceed 2.0 percent when tested as specified in 4.4.

3.6 Shrinkage. Classes 2 and 3 cloths shall be preshrunk and shall not shrink more than 2.0 percent in the direction of the warp or more than 2.0 percent in the direction of the filling when tested as specified in 4.4. The preshrinking process used shall not be identified by name or trademark, either on the cloth, ticket, or package.

3.7 Seam efficiency. The cloth for all types and classes shall have a seam efficiency of not less than 80 percent when tested as specified (see 4.4).

3.8 Length. Unless otherwise specified (see 6.2), the cloth shall be in one continuous length of not less than 40 yards and shall be put up in flat folded pieces as specified in PPP-P-51.

3.9 Fiber identification. Each piece shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under The Textile Fiber Products Identification Act.

3.10 Workmanship. The cloth shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the point level specified.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Examination of the end item. Except when otherwise specified by civil government agencies, examination of the end item shall be in accordance with the provisions of MIL-STD-105 and 4.2.1 through 4.2.4.

4.2.1 *Yard-by-yard examination.* The required yardage of each piece shall be examined on the face side and all defects as defined in Fed. Std. No. 4 which are clearly noticeable at normal inspection distance (about 3 feet) shall be scored and assigned demerit points as listed in 4.2.1.1, except that spots, stains, and streaks shall be counted regardless of cause, i.e., not restricted to rust, oil, or grease, and only those slubs and knots which exceed the maximum acceptable limits shown for each shall be scored (see 4.2.1.2). Defects on the selvage shall not be scored unless the body of the cloth is affected. No yard shall be penalized more than four points. For purposes of this examination, a yard shall begin at the last scored defect. The sample unit for this examination shall be one linear yard. The

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sample size shall be 750 linear yards, regardless of lot size. The sample yardage shall be examined from 25 pieces taken from 25 containers. An approximately equal number of yards shall be examined in each piece of the sample.

The lot shall be unacceptable if the points per hundred square yards exceeds the following values:

Class 1—45.00 points.
Class 2—45.00 points.
Class 3—35.00 points for dyed cloth
42.00 points for bleached cloth

Point computation shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Sample size (yards)} \times \text{contracted width of the material}} = \text{Points per 100 square yards}$$

4.2.1.1 *Demerit points.* Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension — one point

For defects exceeding 3 inches, but not exceeding 6 inches in any dimension — two points

For defects exceeding 6 inches, but not exceeding 9 inches in any dimension — three points

For defects exceeding 9 inches in any dimension — four points

NOTE: The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth

Objectionable odor

Width less than specified

Poor dye penetration, mottled, streaky, or cloudy

Excessive neppiness

4.2.1.2 *Limits of slubs and knots.* Only slubs and knots which exceed the limits shown on figure 1 shall be scored.

4.2.2 *Examination for length.*

4.2.2.1 *Individual pieces.* During the yard-by-yard examination, each piece in the sample shall be examined for gross length. Any gross length found to be less than the minimum specified or more than two yards below the gross length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.2.2.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual gross lengths of pieces in the sample is less than the total of the gross lengths marked on the ticket.

4.2.3 *Examination for shade.* During the yard-by-yard examination, each piece in the sample shall be examined for shade. Any piece in the sample off shade, shaded side to side, side to center or end to end shall be cause for rejection of the entire lot represented by the sample.

4.2.4 *Examination for identification of preshinkage process and non-compliance with the Textile Fiber Products Identification Act.* During the yard-by-yard examination, each piece in the sample shall be examined for these defects. The lot shall be unacceptable if two or more pieces in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

4.3 Examination of preparation for delivery requirements. An examination shall

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be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be L-4 and the AQL shall be 4.0 defects per 100 units.

Examine	*Defects*
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application. Preshrinkage process identified by name or trademark on container.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as: incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Weight	Gross weight exceeds requirements.

4.4 **Testing of the end item.** The methods of testing specified in CCC-T-191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 2½ continuous yards, full width, of the class 1 cloth and 4½ yards full width of the classes 2 and 3 cloths. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. The lot size shall be expressed in units of 1 yard. The sample size (number of sample units) shall be in accordance with the following:

Lot size (yards)	*Sample size*
800 or less	2
801 to 22,000	3
22,001 and up	5

TABLE II.—*Tests*

Characteristic	Requirement reference	Test method
Weight	Table I	5041
Yarns per inch	Table I	5050
Weave[1]	Table I	Visual
Breaking strength	Table I	5100
Seam efficiency[2]	3.7	5110
Colorfastness to:		
Light[3]	3.3.2	5660
Laundering	3.3.2	5610
Wet-dry cleaning	3.3.2	5622
Crocking	3.3.2	5651
Perspiration[4]	3.3.2	5680
Sulfur content	3.3	4.4.1
Nonfibrous material content	3.5	2611
Shrinkage in laundering (percent)[5]	3.6	5550

[1] One determination shall be made from each sample unit and the result shall be reported as "pass or fail."

[2] The thread size as specified in method 5110 shall conform to V-T-276.

[3] In the event of a dispute resulting from the test with method 5660 or as a result of suspected anomalous behavior of certain dye types or formulations, the contracting officer may authorize the exposure to natural light in accordance with method 5662.

[4] The transfer cloth shall consist of a warp of polyester fiber or Orlon 81 and a 6-fiber filling, with equal bars of wool, spun nylon, spun rayon, spun silk, spun acetate, and bleached cotton. Each 6-fiber repeat shall measure approximately 2 inches and shall be separated by a waste filling stripe.

[5] The specimens shall be selected from the cloth (sample unit) as follows: One specimen from each side of the cloth to within three inches of selvage and one specimen from center of the cloth. No two specimens shall contain the same filling yarns. Each elongation value shall be prefixed with a minus sign and both inclosed in parenthesis.

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4.4.1 *Test for sulfur content.*

4.4.1.1 *Solution (acid stannous chloride).* Dissolve 100 grams of stannous chloride crystals in 100 ml. of hydrochloric acid (35 percent concentration) and add 50 ml. of distilled water.

4.4.1.2 *Procedure.* One to 1.5 grams of the finished cloth, cut into small pieces, shall be used for the test. Place the sample in a test tube of suitable size and cover with the stannous chloride solution. Place a filter paper which has been moistened with one drop of lead-acetate solution (made by dissolving 5 grams of reagent-grade lead acetate to 100 ml. of distilled water; add a drop of glacial acetic acid to clear up if solution is cloudy) over the mouth of the tube (punch needle hole in paper). Heat the test tube slowly over a low flame until the solution is boiling. Solution shall not be boiled for more than 10 to 15 seconds, as prolonged boiling will reduce the sensitivity of the test. A brown or black stain appearing on the paper indicates the presence of sulfur.

4.4.1.3 *Evaluation.* The stained filter paper shall then be compared with that obtained from the standard. If the stained area is equal to or lighter than the standard in the degree of staining, the sample shall be considered satisfactory. If the degree of staining by the sample is in excess of the standard, the sample shall be considered unsatisfactory.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and applications of the various levels of packaging and packing for civil agency procurements.)

5.1 **Packaging.** Packaging shall be level A, B, or C (see 6.2 and 6.2.1).

5.1.1 *Levels A, B, and C.* The cloth, put up in flat folds, shall be packaged in accordance with the applicable requirements for class 3 fabrics of PPP-P-51.

5.2 **Packing.** Packing shall be level A, B, or C (see 6.2 and 6.2.1).

5.2.1 *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirements of PPP-P-51.

5.3 **Marking.** In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-51.

6. NOTES

6.1 **Intended use.** The cloth covered by this specification is primarily intended for use in clothing and equipage items.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein, and procurements should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type and class required (see 1.2.1.)

(*c*) Color of cloth required (see 3.3).

(*d*) Width of cloth required (see 3.4.1).

(*e*) Minimum acceptable length, if other than specified (see 3.8).

(*f*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives, as indicated in PPP-P-51 may be exercised in the preparation for delivery of this cloth.

6.3 **Standard sample.** For access to standard sample, address the procuring office issuing the invitation for bids.

6.4 **Transportation description.** The trans-

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portation descriptions and minimum weights applicable to this commodity are:

Rail:

Cotton piece goods, finished, not otherwise indexed by name, in the original piece.
Carload minimum weight — Any quantity.

Motor:

Cotton piece goods, not otherwise indexed, in the original piece.
Motor volume minimum weight — Any quantity.

6.5 Level B packaging and packing shall be used by ordering agencies only when shipment is known to be made to a holding Supply Depot or Warehouse from which partial shipments will be made to ultimate consumers.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—Q M

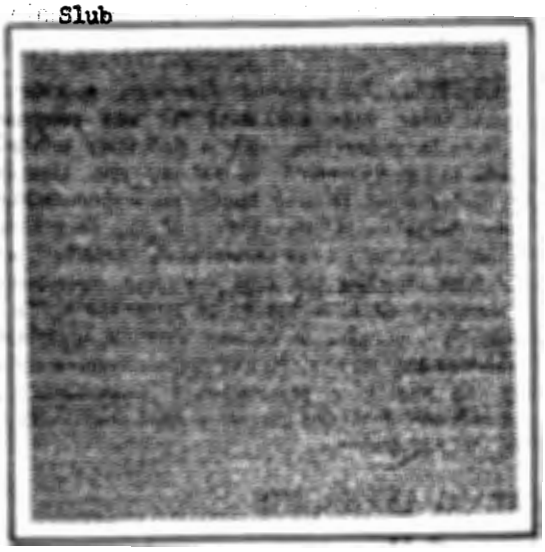
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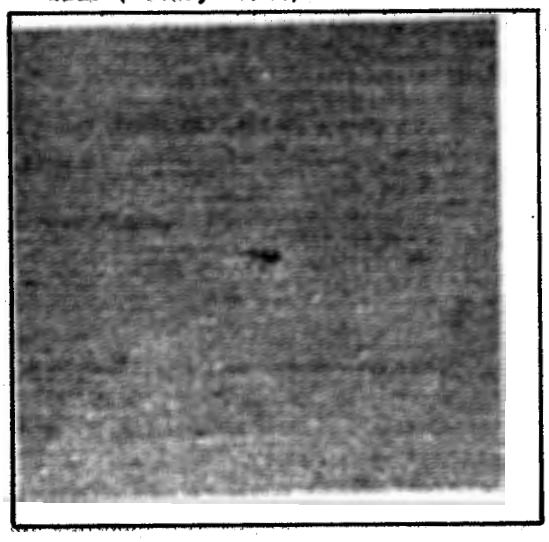
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Excessive neppiness;

Slub



Slub (Loose, Thick)



Lump or Knot (Fuzz)

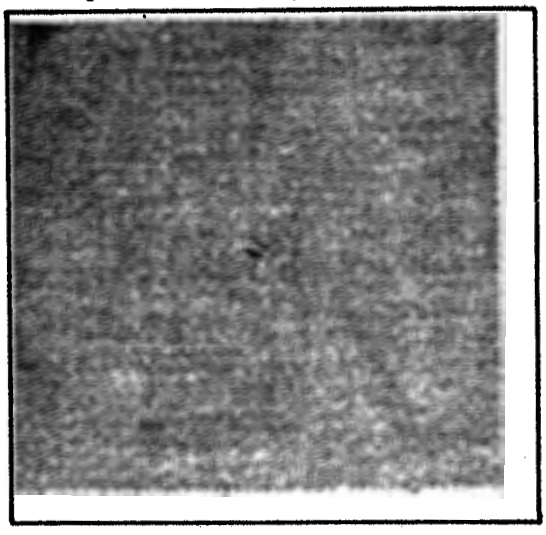


FIGURE. 1.—Limits of acceptability.

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